

Authenticated Image Encryption with Frequency-Domain Convolution Whitening and a Password-Derived Key

M.C. Zuriel López Sosa, Dra. Bárbara Emma Sánchez Rinza, and Dr. Carlos Ignacio Robledo Sánchez

Benemerita Universidad Autonoma de Puebla
zuriel.lopezs@correo.buap.mx, brinza@fcfm.buap.mx, crobledo@fcfm.buap.mx

Source code available at:
<https://github.com/zuriells/cifrado-imagenes-fft>

Abstract. Image encryption is essential for secure digital communication and the protection of visual data. This paper presents a method that combines a frequency-domain *whitening* transform with **authenticated encryption**. The whitening transform relies on the convolution property of the Fourier Transform (FT): the original image and a password-derived key are transformed into the frequency domain and multiplied element-wise, producing a decorrelated representation with no visual resemblance to the original. Since this convolution is a *linear* operation and, by itself, does not constitute a secure cipher, the whitened representation is passed through a **confusion-diffusion** stage (key-dependent permutation and nonlinear diffusion over several rounds) that breaks this linearity and gives the image cipher resistance to plaintext attacks. Finally, confidentiality and integrity are delegated to **ChaCha20-Poly1305** (RFC 8439): the result is sealed with a key derived from the password using PBKDF2-HMAC-SHA256 and SHAKE-256. The result is stored as a 16-bit-per-channel PNG with the nonce, authentication tag, salt, and normalization metadata embedded in the file, removing the need for an external key file. To evaluate the method, over 18 images of different resolution and content from the USC-SIPI database, statistical analyses (histograms, entropy, pixel correlation), a comparison with AES-256, a performance evaluation, diffusion tests (NPCR, UACI, and key sensitivity), and security tests against incorrect-password, Gaussian noise, and resizing manipulations were performed. The results show that the encrypted image exhibits near-maximal entropy (≈ 8 bits/pixel), a uniform histogram, and near-ideal NPCR and UACI values, that decryption with the correct password recovers the original image **exactly** (infinite PSNR), and that **any** alteration of the encrypted file or the password is detected and rejected by the authentication tag, so that the scheme is non-malleable.

Keywords: Image encryption, authenticated encryption, ChaCha20-Poly1305, confusion-diffusion, NPCR, UACI, Fourier Transform, key derivation, PBKDF2, SHAKE-256, integrity, cryptographic analysis.

1 Introduction

In the digital age, information security is a priority, particularly regarding the protection of images, which often contain sensitive data. Digital images, due to their visual nature, present an additional challenge for traditional spatial-domain encryption methods. Frequency-domain processing techniques, such as those based on the Fourier Transform (FT), have gained attention for their ability to effectively decorrelate and conceal data patterns [1, 2]. Existing alternatives include random phase encoding schemes in the optical domain [3] and chaos-based schemes [4, 5], which pursue similar randomization goals through nonlinear spatial-domain dynamics.

However, a frequently underestimated point is that linear combination in the frequency domain — multiplying the FT of the image by that of a key — is itself a *linear* cipher and therefore insecure: given a single (image, ciphertext) pair, an attacker can solve for the key in the frequency domain, $\mathcal{F}(g) = \mathcal{F}(f * g) / \mathcal{F}(f)$, and decrypt any other image protected with the same key. This is precisely the weakness that breaks linear optical double random phase encoding schemes under known- and chosen-plaintext attacks [7, 8]. Consequently, a scheme that rests its confidentiality solely on this operation cannot satisfy the formal security notions, which require that an attacker without the key be unable to distinguish the ciphertext from random noise or to alter it undetected [23, 15].

This work adopts a layered design that resolves this limitation. First, frequency-domain convolution is retained as a **reversible whitening transform** — useful for decorrelating the image and analyzing its statistical properties. Second, a **confusion-diffusion** stage, in the spirit of Fridrich’s architecture [16] (position permutation and nonlinear diffusion over several rounds), breaks the linearity of the whitening and gives the image cipher full diffusion and resistance to known- and chosen-plaintext attacks, evidenced by near-ideal NPCR and UACI values. Third, confidentiality and integrity are delegated entirely to a standard **authenticated cipher**, ChaCha20-Poly1305 [14]. The key is derived from a **password** using standard cryptographic functions (following Kerckhoffs’s principle [9]), so that only the encrypted file and the password are needed to recover the original image.

The aim of this paper is to present and evaluate this method, examining its effectiveness and security. Evaluation is conducted through statistical tests (histogram comparison, entropy, and PSNR) and integrity tests against manipulations of the encrypted file.

The structure of this article is as follows: Section 2 describes the proposed algorithm, detailing the whitening transform, the key derivation, and the encryption and decryption processes. Section 3 presents the tests performed and the analysis of results. Finally, Section 4 concludes with a discussion of the findings.

2 Methodology

The method consists of three clearly separated layers: a **whitening transform** in the frequency domain (reversible, but which does *not* provide security by

itself), a **confusion-diffusion** stage (which breaks the linearity and diffuses any change across the whole array), and an **authenticated encryption** layer (which provides confidentiality and integrity with standard backing). This separation is deliberate: the security guarantees do not depend on the linear operation, but on a nonlinear stage with full diffusion and on a standardized, widely analyzed cryptographic primitive.

2.1 Whitening transform by convolution

The whitening transform is based on the **Fourier Transform convolution property**, which states that the Fourier transform of the convolution of two functions equals the product of their respective Fourier transforms:

$$\mathcal{F}(f * g) = \mathcal{F}(f) \cdot \mathcal{F}(g)$$

Isolating the convolution term and applying the Inverse Fourier Transform ($\mathcal{F}^{-1}$) on both sides gives:

$$f * g = \mathcal{F}^{-1}(\mathcal{F}(f) \cdot \mathcal{F}(g))$$

This implies that convolution in the spatial domain can be computed indirectly through multiplication in the frequency domain, followed by the Inverse Transform. In practice, both transforms are computed using the Fast Fourier Transform (FFT) algorithm [6], which reduces the computational complexity from $O(N^2)$ to $O(N \log N)$.

Throughout this article, f denotes the **original image** and g denotes the **key image**. The whitened representation is the convolution $f * g$, computed in the frequency domain as described above. To invert the whitening, it suffices to solve for f by dividing in the frequency domain by $\mathcal{F}(g)$ instead of multiplying:

$$f = \mathcal{F}^{-1}\left(\frac{\mathcal{F}(f * g)}{\mathcal{F}(g)}\right)$$

Since the images used are in RGB format, all operations are performed independently on each of the three color channels (Red, Green, and Blue).

It must be emphasized that this transform is *linear* and is therefore **not** used as a confidentiality mechanism: it serves only to decorrelate the image before encrypting it. Security is provided by the confusion-diffusion stage (Section 2.3) and the authenticated cipher (Section 2.4).

2.2 Deriving the key from a password

Instead of requiring an external key file, all cryptographic material is generated from a password. This design follows Kerckhoffs’s principle [9]: security relies solely on the secrecy of the password and not on that of the algorithm. The derivation uses password-based key derivation functions [11, 12]:

1. A random 16-byte salt and a random 12-byte nonce are generated at encryption time.

2. The password and salt are processed with **PBKDF2-HMAC-SHA256** [10] using 310,000 iterations, producing a **master secret** of 32 bytes. This secret is the ChaCha20-Poly1305 key.
3. From the master secret, the key image g is generated at exactly the image's size: **SHAKE-256** (an extendable-output function from the SHA-3 family [13]), applied to the secret with a domain separator, is expanded to height $\times$ width $\times$ 3 bytes. The separator avoids reusing the master secret directly as the whitening seed.

Key space. The master secret is 256 bits, so the derived-key space is 2^{256} , well above the 2^{128} threshold considered brute-force resistant. The 128-bit salt ensures that the same password produces different keys on each encryption, and the 96-bit nonce guarantees the uniqueness required by ChaCha20-Poly1305. Resistance with respect to the password itself is determined by its entropy and reinforced by the 310,000 PBKDF2 iterations, which make each guessing attempt costly.

2.3 Confusion-diffusion

So that the method has its own robustness as an image cipher and does not rest its security solely on the authenticated envelope, the whitened representation is passed through a confusion-diffusion stage, in the spirit of Fridrich's classical architecture [16]. $R = 3$ rounds are applied; in each round, over the byte array:

1. **Nonlinear diffusion:** a pseudorandom stream (keystream) is added modulo 256 and an accumulated XOR chaining is applied, so that each output byte depends on all previous ones. The combination of modular addition and XOR introduces nonlinearity and a strong avalanche effect.
2. **Confusion:** byte positions are permuted using a key-dependent permutation.

The keystream, the permutation, and the initialization vector of each round are derived reproducibly with SHAKE-256 from (secret, nonce, round), so that decryption regenerates exactly the same material and reverses the rounds in inverse order. By mixing modular addition, XOR, and permutation, the overall transform ceases to be linear: the known-plaintext attack $\mathcal{F}(g) = \mathcal{F}(f * g) / \mathcal{F}(f)$ no longer applies. The effectiveness of this stage is quantified in Section 3.3 through NPCR, UACI, and key sensitivity.

2.4 Encryption process

1. **Inputs:** the original image f and a password.
2. **Derivation:** the master secret and the key image g are obtained as in Section 2.2.

3. **Whitening:** $f * g$ is computed channel-wise in the frequency domain (Section 2.1). Since the convolution result has a much larger value range than a conventional image, per-channel Min-Max normalization to a 16-bit range $([0, 65535])$ is applied; the per-channel minimum and maximum are kept as metadata so the whitening can be inverted without loss.
4. **Confusion-diffusion:** the whitened representation, serialized to bytes, is transformed with the $R = 3$ rounds described in Section 2.3.
5. **Authenticated sealing:** the result is encrypted with **ChaCha20-Poly1305** [14], using the master secret as the key and the generated nonce. The salt, nonce, iteration count, and normalization ranges are passed as *authenticated associated data* (AAD), so they are also protected against tampering even though they travel in the clear.
6. **Storage:** the encrypted output (the same length as the input) is reinterpreted as a 16-bit-per-channel image and stored as a PNG. The nonce, the authentication tag (Poly1305), the salt, and the ranges are embedded as metadata in a JSON-encoded `tEXt` chunk, so the resulting file is self-contained. This file is the **encrypted image**.

2.5 Decryption process

1. **Inputs:** the encrypted image and the password.
2. **Reading metadata:** the salt, nonce, authentication tag, iteration count, and per-channel min/max ranges are read from the file.
3. **Re-derivation:** the process from Section 2.2 is repeated with the same salt, regenerating the master secret and the key image g .
4. **Verification and decryption:** ChaCha20-Poly1305 verifies the authentication tag before decrypting. **If the password is incorrect or the file was altered, verification fails and the process is aborted without producing any image.**
5. **Inverting the confusion-diffusion:** only when authentication succeeds are the $R = 3$ rounds of Section 2.3 undone in inverse order, recovering the whitened representation.
6. **Inverting the whitening:** the 16-bit normalization is reversed and f recovered by dividing in the frequency domain:

$$f \approx \mathcal{F}^{-1} \left(\frac{\mathcal{F}(f * g)}{\mathcal{F}(g) + \epsilon} \right)$$

with $\epsilon = 1 \times 10^{-10}$ to avoid division by zero.

7. **Rounding and clipping:** the result is rounded and clipped to the valid $[0, 255]$ range to obtain the final 8-bit image.

3 Analysis and Performance Evaluation

This section presents the analysis and evaluation of the proposed method. To verify that the observed behavior is not dependent on a single image, the statistical and security evaluation was carried out on a diverse subset of **18 images**

from the USC-SIPI Image Database [19] — of varied content (natural, aerial, and texture, in color and grayscale) and at resolutions of 256×256 , 512×512 , and 1024×1024 — and the results are reported in aggregate (mean $\pm$ standard deviation). For the illustrative figures and the performance evaluation, a 3024×4032 -pixel photograph and a 2250×2250 aerial image are additionally used. The evaluation comprises: visual inspection, statistical analysis (histogram, entropy, correlation), diffusion (NPCR, UACI, key sensitivity), comparison with a standard cipher, integrity testing, and performance evaluation.

3.1 Visual inspection

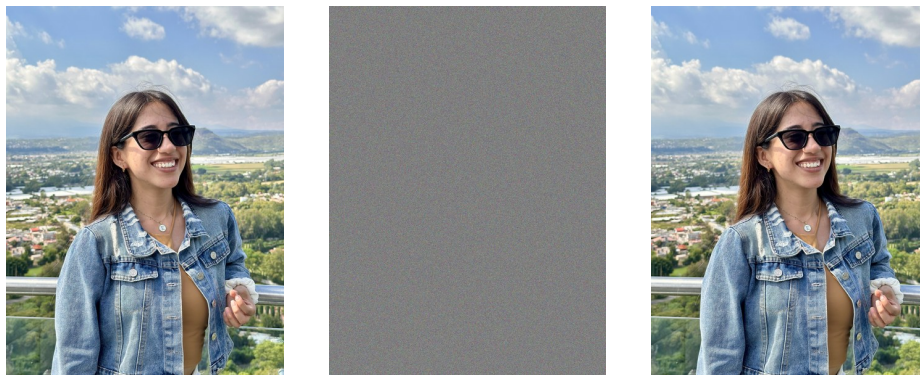


Fig. 1. Left to right: original image f , encrypted image (8-bit preview of the actual 16-bit data), and decrypted image (recovery of f) obtained by decrypting with the correct password.

The encrypted image shows no recognizable trace of f 's content, while the decrypted image is, as quantified in Section 3.2, identical to f .

3.2 Statistical analysis

Histogram analysis. Intensity histograms are a standard tool for analyzing the pixel-value distribution of an image [21]. Histograms of f and of the encrypted image were computed for each color channel (Fig. 2). The original image's histogram shows pronounced peaks associated with regions of uniform color (sky, skin, clothing). The encrypted image's histogram, in contrast, is **essentially uniform**: the peaked structure of the original disappears entirely. This result is consistent with the output of an authenticated cipher, which is computationally indistinguishable from a uniform random sequence. (Note that the intermediate *whitened* representation, before sealing, exhibits a Gaussian shape by the central limit theorem applied to convolution; the uniform histogram in Fig. 2 corresponds to the final, encrypted output.)

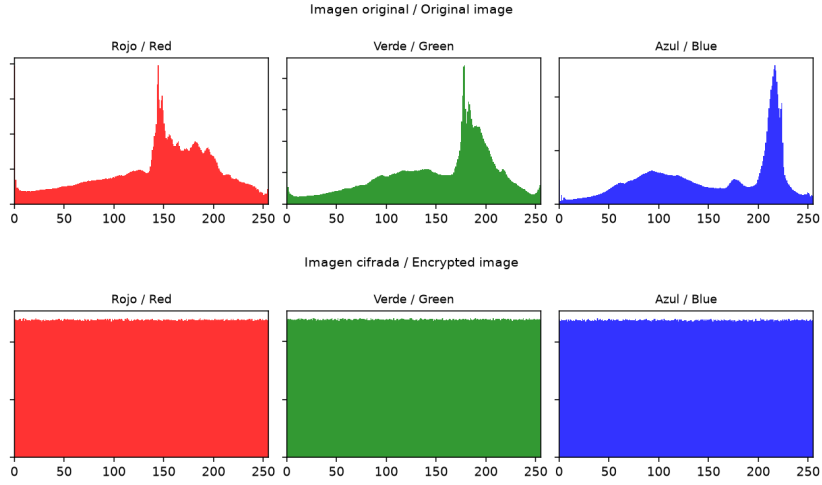


Fig. 2. Per-channel (R, G, B) histograms of the original image (top) and the encrypted image (bottom).

Entropy analysis. Shannon entropy [20] was computed over the *8-bit visualization* of the encrypted image (the high byte of the 16-bit data), averaged over the three channels. Across the 18-image set, the mean entropy of the encrypted image is 7.9990 ± 0.0009 bits/pixel (range [7.9970, 7.9998]), versus 7.0837 ± 0.4082 for the originals. This value is practically equal to the theoretical maximum of 8 bits/pixel and matches that produced by a standard cipher on the same images (AES-256-CTR: 7.9990; Section 3.4). It should be emphasized that this near-maximal entropy stems from the output being that of an authenticated cipher (indistinguishable from a uniform sequence); a plain 8-bit normalization of the whitening, in contrast, concentrates the values into a bell shape and yields substantially lower entropy, and should therefore *not* be used as evidence of randomness.

PSNR evaluation. The Peak Signal-to-Noise Ratio (PSNR) [22] between f and the image recovered by decrypting with the correct password was computed. The result obtained was:

$$\text{MSE} = 0.0 \quad \text{PSNR} = \infty$$

That is, the recovered image is **bit-for-bit identical** to the original. Recovery was exact (MSE = 0, infinite PSNR) for all 18 images in the set, without exception. This confirms that, by using 16 bits per channel with the normalization ranges stored as exact metadata, the full encryption and decryption process introduces no information loss whatsoever.

Adjacent-pixel correlation. In a natural image, neighboring pixels are strongly correlated; a good cipher must reduce this correlation to practically zero [21]. The correlation coefficient between adjacent pixel pairs was measured in the horizontal (H), vertical (V), and diagonal (D) directions, averaged over the channels. Table 1 shows the aggregate values over the 18 images: correlation drops from ≈ 0.88 in the originals to ≈ 0.001 in the encrypted images, in all three directions. Moreover, the global correlation between the original and the encrypted image (plaintext–ciphertext) is 0.0012 ± 0.0012 , i.e., the output retains no linear relationship with the input. Fig. 3 illustrates this effect with adjacent-pixel scatter plots.

Table 1. Adjacent-pixel correlation coefficient (mean $\pm$ standard deviation over 18 images).

Direction	Original	Encrypted
Horizontal	0.9038 ± 0.0817	-0.0014 ± 0.0065
Vertical	0.8979 ± 0.0772	0.0007 ± 0.0103
Diagonal	0.8395 ± 0.1330	-0.0005 ± 0.0070

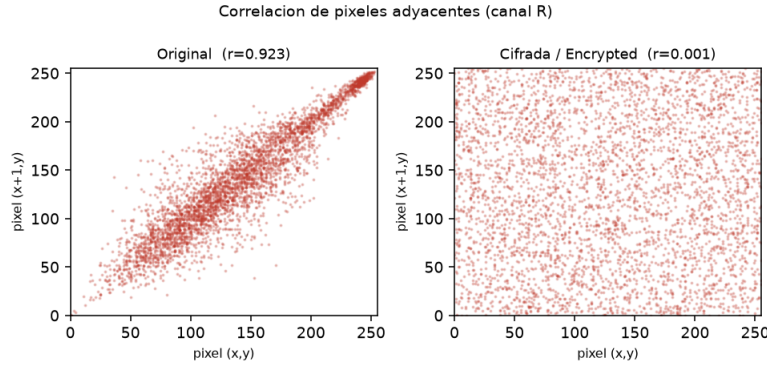


Fig. 3. Adjacent-pixel scatter (R channel, horizontal direction) for a representative image: original (left) and encrypted (right). The strong diagonal of the original disappears entirely after encryption.

3.3 Diffusion: NPCR and UACI

To evaluate the diffusion of the image cipher (the whitening plus the confusion-diffusion stage, before the authenticated envelope), the standard NPCR (Number

of Pixels Change Rate) and UACI (Unified Average Changing Intensity) tests [17] were used, which measure how the encrypted output changes when a single input pixel is modified. For each image, two versions identical except for a single pixel were encrypted with the same key, and the core outputs were compared. Key sensitivity was also measured, by encrypting each image with two keys differing in a single bit. The aggregate results over the 18 images (Table 2) are practically identical to the ideal values, confirming full diffusion: a minimal change in the input, or in the key, alters almost the entire encrypted output.

Table 2. Diffusion metrics of the cipher core (mean $\pm$ standard deviation over 18 images).

Metric	Obtained	Ideal
NPCR	$99.6105 \pm 0.0035\%$	99.6094%
UACI	$33.4658 \pm 0.0173\%$	33.4635%
Key sensitivity	$99.6087 \pm 0.0042\% \approx 99.61\%$	

3.4 Comparison with a standard cipher

To objectively assess the method, its statistical metrics were compared with those of a widely established cipher, AES-256 in CTR mode [18], applied to the same 18 images (Table 4). In both cases the entropy is practically maximal and the adjacent-pixel correlation is indistinguishable from zero, indicating that the output of the proposed method is statistically equivalent to that of a standard cipher. Regarding NPCR and UACI, the proposed core attains near-ideal values, comparable to those reported by chaotic-map image encryption schemes [4, 5]. Note that NPCR and UACI measure diffusion with respect to the *plaintext*, characteristic of image ciphers, and are not applicable to a stream cipher such as AES-CTR (whose security comes from nonce randomization, not per-pixel diffusion).

Table 3. Comparison of statistical metrics (mean over 18 images) between the proposed method and AES-256-CTR.

Metric	Proposed AES-256-CTR		Ideal
Encrypted entropy	7.9990	7.9990	8
$ r $ adjacent (mean)	0.0065	0.0073	0
$ r $ plaintext-ciphertext	0.0012	—	0
NPCR	99.61%	n/a	99.61%
UACI	33.47%	n/a	33.46%

3.5 Integrity and authentication

Unlike a pure linear cipher, in which tampering with the file yields a degraded but potentially still useful image for an attacker, an authenticated cipher is **non-malleable**: any alteration of the password or the encrypted file invalidates the Poly1305 tag, and decryption is **rejected**. Three families of manipulations were evaluated — including resizing with symmetric factors ($\pm 1\%$, $\pm 5\%$, $\pm 10\%$) — plus a single-bit flip; the results are summarized in Table 3. It should be clarified that the inability to decrypt a manipulated image is **not** interpreted here as heuristic “tamper detection,” but stems from an explicit authentication mechanism (the Poly1305 tag), which is what legitimizes the integrity claim.

Table 4. Outcome of the decryption attempt under different manipulations of the encrypted file or the password (identical across all 18 images).

Attack / manipulation	Outcome
Incorrect password	Rejected
Gaussian noise ($\sigma \approx 32.8$)	Rejected
Gaussian noise ($\sigma \approx 163.8$)	Rejected
Gaussian noise ($\sigma \approx 327.7$)	Rejected
Resizing -1% , -5% , -10%	Rejected
Resizing $+1\%$, $+5\%$, $+10\%$	Rejected
Single-bit flip	Rejected

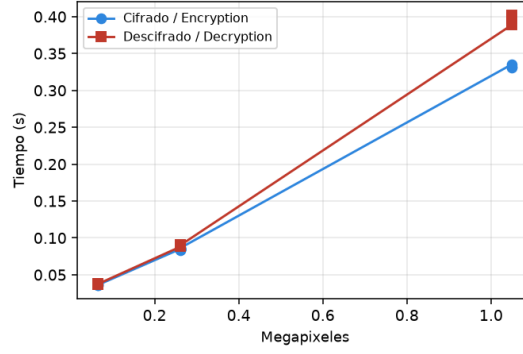
In all cases, including the smallest possible perturbation — a single-bit change — ChaCha20-Poly1305 detected the alteration and rejected the decryption. This behavior is qualitatively superior to that of an unauthenticated cipher: the system not only protects confidentiality but **guarantees the integrity and authenticity** of the encrypted file, so that no manipulation — however small — can go unnoticed or be exploited to obtain a partial reconstruction of the image.

3.6 Performance evaluation

Encryption and decryption times were measured for images of different sizes, from 256×256 up to 3024×4032 pixels (Table 5). Throughput stays around 6–9 MB/s. It is worth noting that the PBKDF2 key derivation (310,000 iterations) is a fixed per-operation cost, deliberately chosen to hinder brute force on the password; it therefore dominates the time for the smallest images. The remaining time grows approximately linearly with the number of pixels (Fig. 4), consistent with the $O(N \log N)$ cost of the FFT and $O(N)$ of the confusion-diffusion. Memory consumption also scales linearly with image size; its peak (dominated by the confusion-diffusion permutations) was about 4.5 GB for the 12-megapixel image, which constitutes a limitation to be optimized in future work (for example, by block-wise processing).

Table 5. Encryption/decryption times and throughput by image size.

Size	MP	Encrypt (s)	Decrypt (s)	Throughput (MB/s)
256×256	0.07	0.05	0.04	4.2–5.3
512×512	0.26	0.09	0.09	9.0
1024×1024	1.05	0.33	0.40	7.9–9.5
2250×2250	5.06	1.92	2.25	6.7–7.9
3024×4032	12.19	5.41	6.13	6.0–6.8

**Fig. 4.** Encryption and decryption times as a function of the number of megapixels.

4 Conclusions

The analysis demonstrates that the proposed method effectively transforms an image’s content into an encrypted representation with a randomized appearance, near-maximal entropy (≈ 8 bits/pixel), and a uniform histogram, computationally indistinguishable from noise. The central point of the design is to recognize that linear convolution in the frequency domain, although useful for decorrelating the image, is insecure on its own — vulnerable to known- and chosen-plaintext attacks [7, 8] — and therefore to add a confusion-diffusion stage that breaks this linearity and achieves full diffusion (near-ideal NPCR 99.61% and UACI 33.46%, with 99.61% key sensitivity), and to delegate confidentiality and integrity to a standard authenticated cipher (ChaCha20-Poly1305).

The evaluation over a set of 18 images of different resolution and content confirms that these results are not image-dependent: entropy 7.9990 ± 0.0009 , adjacent-pixel correlation reduced from ≈ 0.88 to ≈ 0.001 , and plaintext–ciphertext correlation of ≈ 0.001 , values statistically equivalent to those of a standard cipher (AES-256). Decryption with the correct password recovers the original image **exactly** (infinite PSNR) in all of them, thanks to the 16-bit-per-channel storage with precise normalization ranges. The integrity tests show that **any** alteration

of the password or the encrypted file — noise, resizing, or even a single-bit change — is detected and rejected by the authentication tag, a non-malleability property especially valuable for detecting and preventing unauthorized tampering. Overall, the scheme provides confidentiality, integrity, and authenticity with security backed by standard primitives, while also removing the need to distribute a separate key file, making it practical for secure image transmission.

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